User Manual

Outdoor Unit Gateway

V2.1

Qingdao Maisi Intelligent Technology Co.,Ltd.

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Section 1 Product overview

1.1 Product introduction

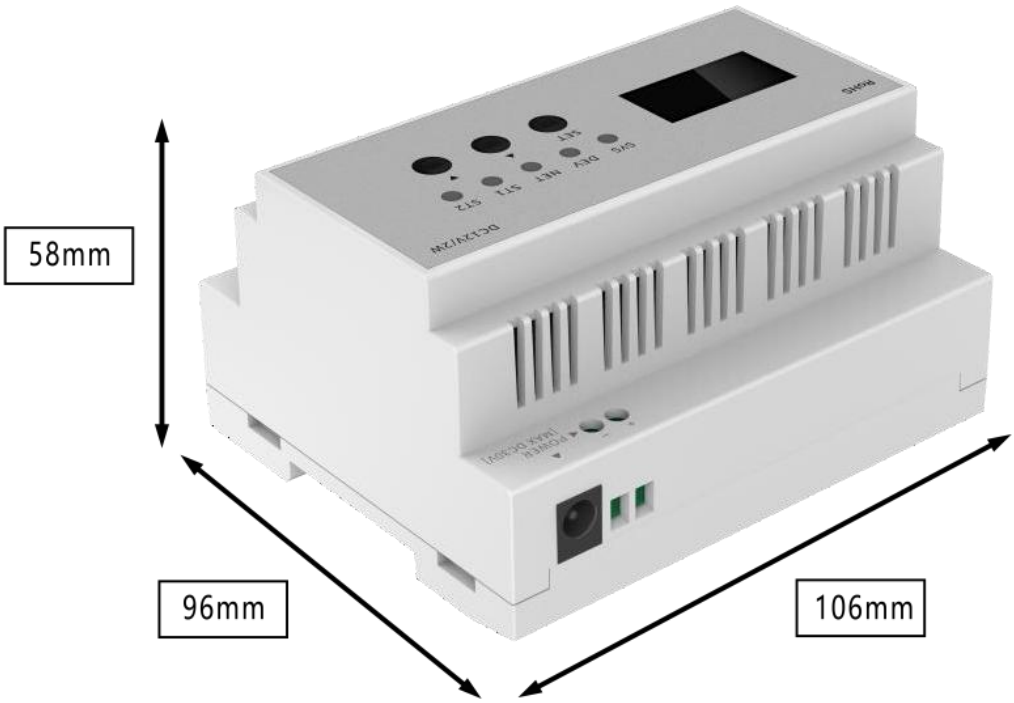
Thank you for choosing Maisi multifunctional Gateway. This gateway can connect the air conditioner to the “smart home centralized control system” or “BMS system”. This gateway can be provided as “RS485” or “KNX” or Third-party wireless module interconnection.

The supporting air conditioner brands include: DAIKIN / Hitachi / TOSHIBA / Mitsubishi Electric / Haier / Hisense /Panasonic / YORK/ Mitsubishi Heavy Industries / GREE /Midea /LG /AUX /BOSCH /Samsung/ CHIGO / TICA / Carrier / TRANE / McQuay.

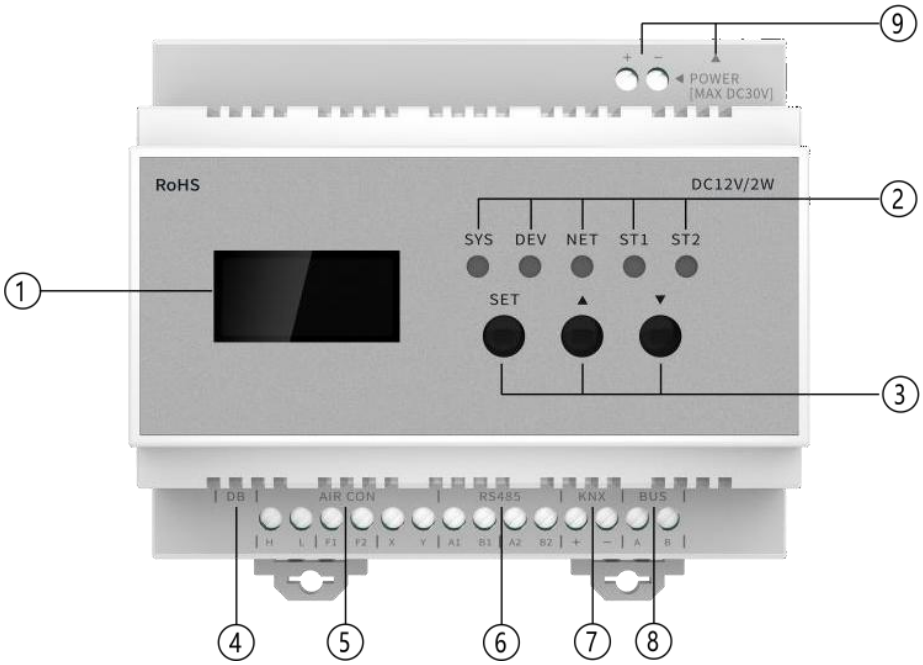
Before installation and use, please read this manual carefully to fully understand the product.

1.2 Product appearance and electrical parameters

1.2.1 Product appearance



Gateway dimension diagram



① **Display screen：** It can display whether the communication is normal, display the air conditioner brand under control, the number of air conditioners, the number of new fans, and the equipment parameters of this product.

② **LED indicator:** Indicator for working status.

③ **Button:** You can set different functions by pressing the button. Specific functions please refer to the details in Button instructions.

④ **DB:** Debug and download programs.

⑤ **AIR CON:** according to the protocols, connecting to DAIKIN / Hitachi / TOSHIBA / Mitsubishi Electric / Haier / Hisense /Panasonic / YORK/ Mitsubishi Heavy Industries / GREE /Midea /LG /AUX /BOSCH /Samsung/ CHIGO / TICA / Carrier / TRANE / McQuay and other air conditioners through different ports.

⑥ **RS485:** This port is for New fan, floor heating controller or other controllers based on RS485 protocol.

⑦ **KNX:** This port is for smart home system and Air conditioner system based on KNX.

⑧ **BUS:** For smart home system and Air conditioner system based on 485 protocol.

⑨ **POWER:** The standard power supply is DC12V. If mismatched power supply is used, the product may be damaged.

1.2.2 Electrical parameters

Items	Specification
Product dimensions	106*86*58mm
Power	DC 12V
Power consumption	2W and below

Work environment	-20-80℃，90%RH
Configure parameters and upgrade	Bluetooth + We chat mini programs
Installation mode	Install on the guide rail or Placement
Port	ZigBee/BLE/WIFI/RS485

1.3 Support protocols and specifications

1.3.1 Support the following protocols:

- MODBUS-RTU
- KNX（need to specify in advance when purchasing）
- RS485（Private protocol）
- Third-party wireless module（Zigbee / BLE / WiFi）

1.3.2 Specification of communication

Items	Parameters
transmission mode	Half duplex
baud rate	9600bps
Starting bit	1 bit
Data bit	8 bit
Checkout bit	Even
Stop bit	1 bit

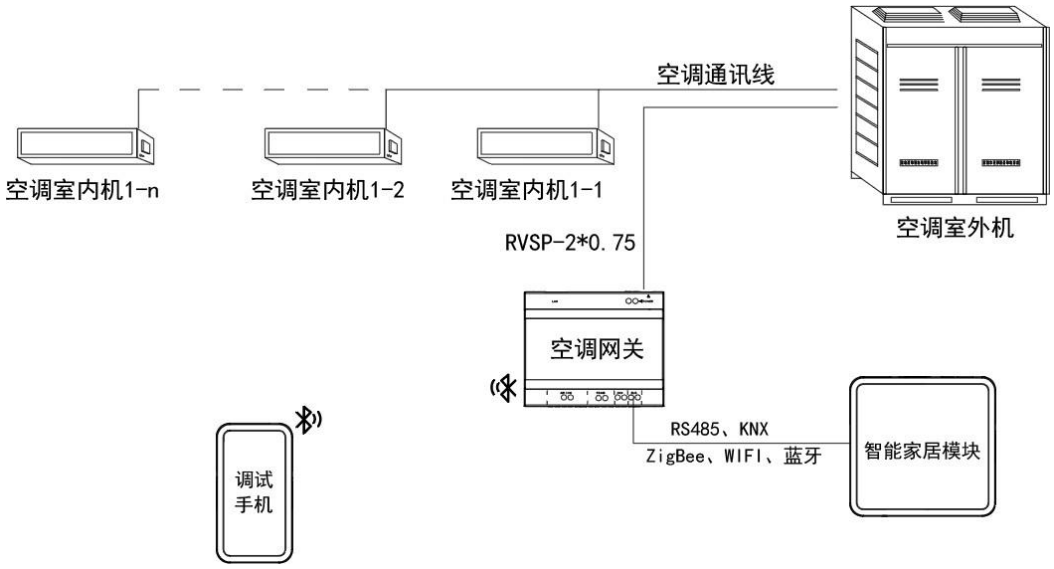
Tips：Parameters can be modified through WeChat mini program.

Section 2 Installation Instructions

2.1 Installation Environment

This product needs to be installed in the weak current control panel with guide rail or Smart home control box and ensure it’s installed in indoor facilities.

2.2 Wiring connection and cautions



The communication line between the air conditioner indoor unit and the outdoor unit is responsible for the air conditioner constructor. Like the refrigerant copper tube between the indoor and outdoor air conditioner, it belongs to the part of the air conditioner system. And it doesn't need the weak electricity integration.

The gateway is connected to the outdoor unit of the air conditioner system through 2 core lines, and then communicates with all the indoor machines in the air conditioner system for monitoring. The shielded twisted pair line from air conditioner outdoor gateway to this gateway requires more than 0.75mm2 line diameter, the total length is not more than 1000 meters, which can be reserved by the air conditioner constructor or in the later period. The communication between the gateway and the weak current integrated system is also realized by twisted pair or cable. The parameter requirements and construction details of the communication line are determined according to the location between the module and the weak current integrated system.

The gateway needs to be installed in the household, which can be installed in the smart home control cabinet or near it.

Notice of connection operation:

1) Different control modes of terminal terminals are different

2) It is recommended to use the power supply adapter packed with the device or DC12V power supply adapter produced by a qualified manufacturer.

3) Wrong wiring may damage the device. If the power cable is connected to the signal terminal, it will cause a fault. Please contact the manufacturer. Do not disassemble and maintain it without permission.

4) The power supply of gateway and air conditioner must be cut off during wiring operation.

5) All signal cable cables do not need to have side-by-side connection with the power line

and other signal lines, which may cause misoperation due to interference. If requiring side-by-side wiring, Please keep the connection distance at 15 cm between the air conditioner, and 30cm distance with other wiring, or connect to the iron protection tube and make one end of the protective tube grounded. The chassis must be D grounded.

2.3 LED indicator description

Mark	Described device	State	Description
SYS	Gateway	Red light steady on	Unauthorized
		Green light flash slowly	Normal
		Yellow light steady on	Abnormal
		Green light steady on	Initializing
DEV	Air conditioner	Green light steady on	Searching
		Green light flash	Searching complete. usual status：Green light flash slowly； Data transmission：Green light flash fast
		Red light steady on	Search failed； Fault reporting； No data transmission。
NET	Smart home	Green light flash	Module failure（ZigBee、WIFI、KNX）
		Yellow light steady on	Module falling off（ZigBee、WIFI、KNX）
		Green light flash fast	In connection or network Settings
		Green light steady on	After the distribution network or connection is completed, the green light remains on for 1 minute
		Green light flash	Data transmission：Green light flash fast； No data transmission：Green light flash slowly.
		OFF	Unconnected or the network is not configured
ST1	Bluetooth	OFF	Bluetooth disconnected
		Green light steady on	Bluetooth connected
ST2	WIFI	OFF	WIFI disconnected
		Green light steady on	WIFI connected

Section 3 Hardware Configuration

3.1 Hardware configuration preparation

1) Confirm whether the air conditioner system is in power up state.

2) Confirm that the air conditioner system has been successfully debugged by the air conditioner manufacturer, without any alarm or exception;

3) Confirm the address of the air conditioner indoor unit and the corresponding outdoor unit and record it (For setting and query methods, please refer to relevant documents on company website and Wechat official account);

4) Connect the air conditioner communication line and connect KNX, RS485,Zigbee,WIFI, or BLE to the corresponding interface of the gateway as required;

5) Select the port and match the communication parameters for the device.

3.2 Connection port selection and use

Brand	Gateway Port	Air-Conditioning Port	Port polarity	Wiring from the outdoor unit	Centralized control address
Dakin-multi outdoor units	F1，F2	F1,F2 outside-outside	Nil	YES	YES
Dakin-single indoor units	F1，F2	F1,F2 inside- outside	Nil	NO	YES
Hitachi	F1，F2	1，2	Nil	NO	NO
Toshiba	F1，F2	U1,U2	Nil	NO	NO
Mitsubishi Electric	F1，F2	M1,M2 (TB7)	Nil	YES	NO
Mitsubishi Electric(One outdoor unit)	F1，F2	M1,M2 (TB3)	Nil	NO	NO
Mitsubishi Electric(Power Multi)	F1，F2	M1,M2 outdoor unit M1,M2 (switch box)	Nil	NO	NO
York-Qingdao	F1，F2	1，2	Nil	NO	NO
Panasonic	F1，F2	U1，U2	Nil	NO	NO
	F1，F2	U1，U2	Nil	NO	NO
Hisense	F1，F2	1，2	Nil	NO	NO
Haier	F1，F2	P，Q	Nil	NO	NO
Gree	H，L	G1，G2	Yes	YES	NO
	H，L	D1，D2	Yes	NO	NO
MHI	X，Y	A1，B1	Nil	NO	NO
	X，Y	A2，B2	Nil	NO	NO
TRANE	X，Y	A，B	X-A,Y-B	YES	NO
Midea/Carrier/ Bosch	X，Y	X，Y	X-A,Y-B	YES	NO
	X，Y	P，Q	P-A,Q-B	NO	NO
Aux	X，Y	X，Y	Ye	NO	NO

Midea-Coolfree	X，Y	A，B	X-A,Y-B	NO	NO
LG	X，Y	A、B（CEN.)	X-A,Y-B	NO	YES
York-ATW	X，Y	A、B	X-A,Y-B	YES	NO
	X，Y	A2、B2	X-A2,Y-B2	YES	NO
McQury-ATW	X、Y	Apc、Bpc	X-ApcY-Bpc	YES	NO
Carrier-ATW	X、Y	A、B	（X-A,Y-B）	YES	NO
TRANE-ATW	X、Y	CN29(A、B)	（X-A,Y-B）	YES	NO

Note 1:

If the port polarity is Nil, the wiring connection between gateway port and air conditioner port can interchange; if the port polarity is Yes, the wiring connection between gateway port and air conditioner port must match the ports between them strictly following the specification. For example: Midea, the gateway port is “X, Y” and the air conditioner port is “X, Y”. When the port polarity is Yes, it needs to connect the Port X of air conditioner to the Port X of gateway and the Port Y of air conditioner to the Port Y of gateway.

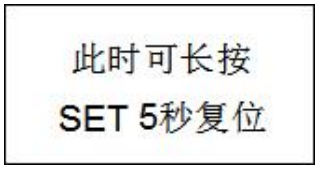
Note2:

Connect TB7 "M1,M2" to the centralized control cable when Mitsubishi electric has multiple outdoor units, and plug the jumper cap of the outdoor PCB board centralized control switch CN41 into the CN40.

3.3 Display description

1.After the gateway is powered on, the "Gateway Bluetooth Configuration" mini program QR code is displayed. Scanning this QR code can enter the WeChat mini program to upgrade the firmware of this gateway, as shown in the following figure:

2. After the QR code disappears, the LCD screen displays: "此时可长按 SET5 秒复位”. If reset operation is not required, wait for automatic skip; If a reset is required, when the display reads "At this time, press and hold SET for 5 seconds to reset", press the "SET" button for about 5 seconds, and the LCD screen displays "Reset successful" to complete the reset.



3. At this point, you will enter the information page, where the first page displays the air conditioning brand, supported protocols, number of air conditioners found, and number of air conditioners in operation/error reporting (when the air conditioning quantity is 0, it is searching for air conditioners, and you need to wait for about 5 minutes to refresh the

interface to display the number of air conditioners found). At this time, press "▼" to switch to the next page, where the first line displays the local IP and the second line displays the firmware version number. At this point, press "▼" again to switch to the next information page, displaying the presence or absence of remote functions, brand information of fresh air and floor heating, etc. Press "▼" again to switch to the next information page, displaying local and module fault information, etc. Press "▼" again to switch to the next information page, displaying WIFI, RSSI, Bluetooth and Bluetooth addresses, etc. Each information page can be freely switched through "▲" and "▼", as shown in the following figure:

3.4 Button description

SET：After the device is powered on again, hold down this button to restore the factory defaults。

Combo buttons：

SET+UP:（search completed）Press and hold for 5s to turn on all air conditioners.

SET+DOWN:（search completed）Press and hold for 5s to turn off all air conditioners.

UP+DOWN: The gateway restart.

Section 4 Bluetooth configuration

4.1 configurate description

By connecting to the gateway through Bluetooth, you can modify or upgrade the basic information of the gateway, system parameters, air conditioner brand, integrator, etc. You can also control the air conditioner connected to the gateway. The specific connection method is as follows:

Search for the official account of”迈斯" or "迈斯 maisi" in the wechat search bar. After following the official account, click "常用工具” → “蓝牙配置" below to enter the Bluetooth configuration interface to turn on the Bluetooth of the mobile phone and select the gateway device to be connected.

After Bluetooth is enabled on the mobile terminal, click the unconnected device, select the device, and connect the mobile phone to Bluetooth. After successful connection, you can access the gateway webpage.

4.2 Webpage for gateway:

This page contains basic information, status information, gateway Settings, you can view the basic information and status information of the gateway, and the gateway network Settings, WIFI Settings, remote, KNX, RS485, air conditioning Settings, gateway restart.

Basic information: including equipment model, SN code, controlled air conditioner brand, product software version, integrator information, controlled fresh air and controlled floor heating.

Status information: includes gateway authorization status and product fault information.

Gateway Settings: including network Settings, WIFI Settings, remote Settings, KNX Settings, RS485 Settings.

4.3 Webpage for air conditioning

This page includes air conditioning setting, host setting, group setting, fresh air setting, floor heating setting, group setting, switch, temperature adjustment, wind speed adjustment and other Settings for HVAC equipment. For example, air conditioning setting is as follows:

4.4 Webpage for upgradation

You can upgrade the gateway software on the upgrade page based on the brand of the air conditioner and fresh air device to be connected.

4.5 Webpage for Debugging

The debug page is for the use of developers only.